MMCS Group Project · 2026 · Tier 1

Recovering the value of surplus renewable generation

Battery or electrolyser, at what scale, and under which contract

PREPARED BY	VERSION	DATA WINDOW	DECISION OWNER
Group 15	1.0 — 5 November 2026	1 Jan 2020 – 17 Sep 2026	Portfolio board

HEADLINE RESULTS

+£8.57m	20 MW	2.6 yr	0.07%
annual value of the 20 MW electrolyser module	electrolyser; no battery is economic	payback before stack replacement	of current electricity sales diverted to hydrogen

Decision

Build **20 MW of PEM electrolysis** and **no battery**, conditional on a hydrogen offtake of at least 2.2 t/day at £7.80/kg *and* on the operating mandate to divert 0.07% of current electricity sales. Capital is £29.2m, the annual increment £8.57m, the payback 2.6 years before stack replacement. Under a current-generation-first obligation the same asset loses £1.07m a year, so the mandate — not the price — is the binding condition.

1 Executive summary

The client's portfolio generated 215.7 TWh in 2023 against 245.8 TWh of demand: 87.5% of demand can be served, and 549.9 GWh of generation is stranded on the 29 days when supply exceeds demand. Of the two candidate assets, the brief's 1.5 MW/3 MWh battery is not economic at any non-negative capital cost, and a 20 MW electrolyser adds £8.57m a year — positive in every year replayed from 2020 to 2026, weakest in 2021 at £6.29m.

The mechanism is not the surplus but the contract. The module produces at most 8.59 t/day of hydrogen, just under the buyer's 10 t/day cap, so capacity is set by the offtake ladder rather than by the stranded energy. Of the 171.8 GWh the electrolyser consumes, 13.9 GWh (8.1%) is electricity that would otherwise be unused and 157.9 GWh (91.9%) is diverted from direct sale: the honest headline is that eight per cent is avoided waste and ninety-two per cent is an alternative use of saleable electricity.

What the reader has to accept: daily averages with perfect foresight and constant 2020 money. The number that would move most if it failed is the offtake contract, not the capital cost — moving CAPEX across the whole £0.82m–£1.65m/MW range changes the answer by at most £1.1m a year, while losing the operating mandate takes +£8.57m to £0.

The assumption that carries the answer

The recommendation holds while the client may divert about **0.07% of current sales** (157.9 GWh). A 99.9% supply floor still permits the full £8.57m; a 99.99% floor permits 35 GWh of hydrogen and £1.04m; a 100% floor excludes the investment entirely.

2 Problem, boundary and data

2.1 The question and the arithmetic

The decision is how much battery and electrolyser capacity is worth adding, and how much otherwise unused electricity it captures. Three numbers frame it: 215.7 TWh generated, 245.8 TWh demanded, 549.9 GWh stranded on 29 days. The first two say the portfolio is already short of its own demand, so any new load is an alternative use of saleable electricity; the third says how much genuinely wasted energy is available to compete for. That ratio — 0.26% of one year's generation — is the ceiling on the physical prize.

2.2 Boundary and conventions

We use the five supplied tables in the independently repaired release (2,452 days, 1 January 2020 to 17 September 2026), keeping the course originals as a sensitivity: the original price is about half the repaired value and the original demand is truncated on the last day of each weekly query. Supplied generation plus the specified wind farm is a *constructed* system, not observed curtailment, and demand is the absorption cap — no purchases, no additional export route, no curtailment payment. The analysis is daily and perfect-foresight, without network, intraday, ancillary or hydrogen storage; money is constant 2020 GBP at 5% real; the battery is the brief's 1.5 MW/3 MWh module (90% round-trip, 20–80% window); the electrolyser is 20 MW PEM at 55.9 kWh/kg; hydrogen is valued at £7.80/kg in 2020 money, close to the brief's £9.5/kg in 2023 money (£7.91).

3 Model

For day t, with generation G_t, demand D_t and market price p_t, the model chooses direct sale q_t, battery charge c_t and discharge d_t, electrolyser input h_t, unused energy u_t, operating modules z_t and installed units n_B, n_H:

$$\max\ \Pi = \frac{1}{Y}\sum_t \left[\frac{p_t\,(q_t + d_t)}{1000} + \frac{v\,h_t}{55.9}\right] - a_B\,n_B - a_H\,n_H \tag{1}$$

subject to

$$q_t + c_t + h_t + u_t = G_t, \qquad q_t + d_t \le D_t, \tag{2}$$

$$0.096\,z_t \le h_t \le 0.48\,z_t \le 0.48\,n_H, \qquad h_t \le 55.9\,L/10^6, \tag{3}$$

$$\textstyle\sum_t d_t = 0.9\sum_t c_t, \tag{4}$$

where p_t is £/MWh, v is the hydrogen value in £/kg, $\eta = 55.9$ kWh/kg, L is the offtake cap in tonnes per day, a_B, a_H are annualised capital charges (15-year battery, 30-year electrolyser, fixed O&M and stack replacement at years 11 and 22) and Y is the number of days in the year. Policy A lets the model allocate each day's generation economically; policy B adds $q_t = \min(G_t, D_t)$, so current generation serves current demand first. Policy B is the operating reality we test against, not a modelling preference.

The annual battery balance (4) is the Tier 1 time simplification and is an upper bound on battery value; no selected configuration uses a battery, so it cannot change the recommendation. One design year has 2,556 rows and 2,192 columns, solves to proven optimality in under 0.1 s with Xpress 9.9.1, and 185 optimisation cases plus 2 rule-based comparators were run; twenty independent checks pass.

4 Results

Table 1. Configuration comparison, 2023, annual increment versus the matching no-investment case, £m/yr.

Policy	No invest.	Battery	H_2 only	Rule-based
A: economic allocation	0	0	**+8.57**	−1.07
B: current generation first	0	0	0	−1.07

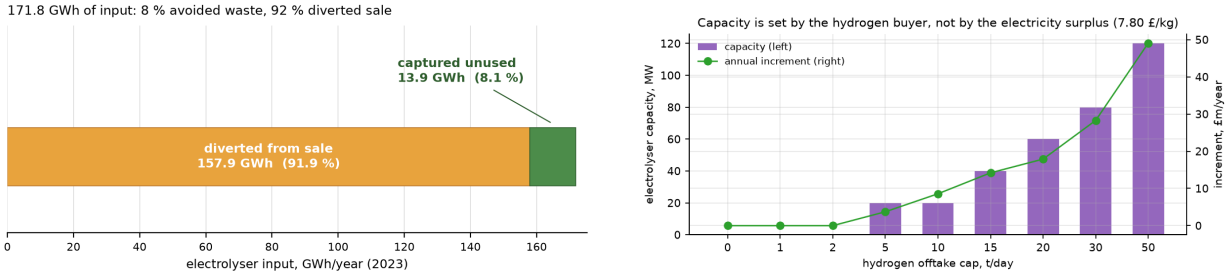
"Both" equals the electrolyser case because the battery is never selected; zero under B means the asset is not run.

Table 2. Where the £8.57m comes from, 2023, £m/yr.

Line	£m/yr	Note
Hydrogen revenue	23.97	3,074 t over the year at £7.80/kg
Electricity revenue forgone	−12.39	157.9 GWh at the daily price
Fixed O&M; stack reserve; capital charge	−3.01	0.52 + 0.59 + 1.90
Net annual increment	**+8.57**	30-year NPV +£132m

The NPV is a scenario value, not a forecast: no price decay, no degradation, no terminal value.

The battery is never selected. At zero capital cost and zero fixed O&M a 1.5 MW unit is worth £13.25k a year, while the primary-scheme fixed O&M alone is £22.3k per unit, so no unit is economic at any non-negative capital cost; on the lower table-based O&M the break-even capital cost is £7.0k/MWh, about 37 times below the brief's €288k/MWh (£256k/MWh). Because the annual-pool model overstates battery value, the conclusion is robust to the relaxation.



(a) attribution of electrolyser input (b) capacity selected against the offtake cap

Figure 1. Left: the 171.8 GWh electrolyser input splits into 13.9 GWh of avoided waste and 157.9 GWh diverted from sale. Right: the capacity ladder is driven by the contract, not by the surplus.

Source: `results/results.json`, package `tier1/20261105-v1`; panel (a) redrawn from the package figure for legibility, same data.

On the 29 surplus days the renewable share of generation is 52.7%, so a pro-rata attribution of the captured energy is about 7.3 GWh; at daily resolution the strict renewable bound is 0 to 13.9 GWh. The capacity ladder follows the contract: 0 MW at 2 t/day or less, 20 MW at 5–10 t/day, 40 MW at 15, 60 MW at 20, 80 MW at 30 and 120 MW at 50 t/day (+£49.1m a year on £175m of capital). At a 10 t/day cap the investment switches on at a netback of £4.62/kg; at £7.80/kg it needs 2,230 kg/day, and at £9.5/kg, 1,717 kg/day. Fixing 20 MW and replaying each year keeps policy A positive throughout, from +£6.29m (2021) to +£16.72m (2020), while policy B loses money in 2023 and 2024 (−1.07m and −1.00m). The spread is business risk, and it is the freedom to allocate generation that the asset depends on.

5 What changes the recommendation

Electrolyser CAPEX between £0.82m and £1.65m/MW gives +£9.41m to +£8.33m with capacity unchanged, and a real discount rate of 0% to 12% gives +£9.44m to +£6.99m. The strict 2023 sample gives +£8.69m and removing the wind farm +£8.13m; the unrepaired price would give +£15.19m, which is why it is not used. None of these moves the decision — the supply obligation in the caveat above does.

6 Recommendation

1. No battery in Tier 1 at the brief's cost. Re-open it only with intraday or ancillary revenue, or a capital cost near £7k/MWh.
2. Negotiate the hydrogen contract first: at least 2.2 t/day at £7.80/kg, or 1.7 t/day at £9.5/kg. Below a £4.62/kg netback, do not build.
3. Build 20 MW only with the contract *and* the operating mandate to divert about 0.07% of sales.
4. Stage the expansion: 20 MW for a 10 t/day contract, 120 MW for 50 t/day. Do not pre-build.
5. Report the capture honestly: 13.9 GWh of avoided waste (8% of input) and 157.9 GWh diverted from sale (92%).

7 Validation and limitations

Every stored plan closes its daily energy balance, respects the demand, charge and discharge, module and offtake caps, uses integer units and satisfies the annual battery balance; revenues and costs are recomputed from the plans. The reference cases reproduce the independently built Baseline 1104 package (A0 17,139.332; 20 MW +8.572; B fixed 20 MW −1.065; simple rule −1.065; 2024 +11.951; old price +15.188) — arithmetic agreement between two implementations, not an endorsement of shared assumptions.

Limitations, each measured rather than conceded: daily averages remove intraday value, so the battery figure is an upper bound; the annual battery balance is a relaxation; the model is perfect-foresight; the hydrogen value is a policy-supported reference, not a contract; CAPEX is a modern installed midpoint while O&M and stack replacement use older UK proxies; there is no degradation, hydrogen storage, network constraint or ancillary market; 125 days carry imputed fuel composition (85 in 2023), so the renewable attribution is conditional on the repair; and 2026 is partial.

Reproducibility and references

[1] Data: Group 15 release `full_data_repair/20261104-v2`, 2,452 days, accessed 4 November 2026; repair log in the same release.
[2] Brief and teacher clarifications, 2026: asset specifications, 10 t/day offtake cap, discount convention.
[3] Hydrogen reference: policy-supported strike value, £9.5/kg (2023 money), deflated with ONS CPI series D7BT; cost and O&M tables with vintage in `parameters.json`.

Reproduction: in package `tier1/20261105-v1`, `reproduce.py` reruns all stages, `src/verify.py` asserts 20 checks, `MANIFEST.json` digests 45 files; solver Xpress 9.9.1.